

How to set up ESP32-WROOM-32.

Hey guys, this is going to be a quick guide on how to set up the ESP32-WROOM-32 board. This method is also applicable to some other ESP32 variants, but since I have an ESP32-WROOM-32 board with me, all the board pictures you will be seeing in this tutorial will be of that board.



We will just get straight into it and not bore ourselves with some technical details of the board. You can always read more or check the datasheet of the board to learn more about what you can use the WSP32-WROOM-32 to do or how powerful it is. I assume since you are here, you somehow have an idea of what the board is.

We need to get four things done: set up the Arduino IDE, get the board installed, get the driver installed, and test it out.

Arduino IDE Set-Up

There are lots of software packages that can be used for developing the ESP32 boards; we have the ESP HomeKit SDK, Arduino IDE, etc but the Arduino IDE is by far the easiest to get started with. If you don't already have Arduino IDE installed, kindly download the software from the Arduino website, <https://www.arduino.cc/en/software>. After the download is done, install the software, and you are good to go.



Downloads



Arduino IDE 2.2.1

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger.

For more details, please refer to the [Arduino IDE 2.0 documentation](#).

Nightly builds with the latest bugfixes are available through the section below.

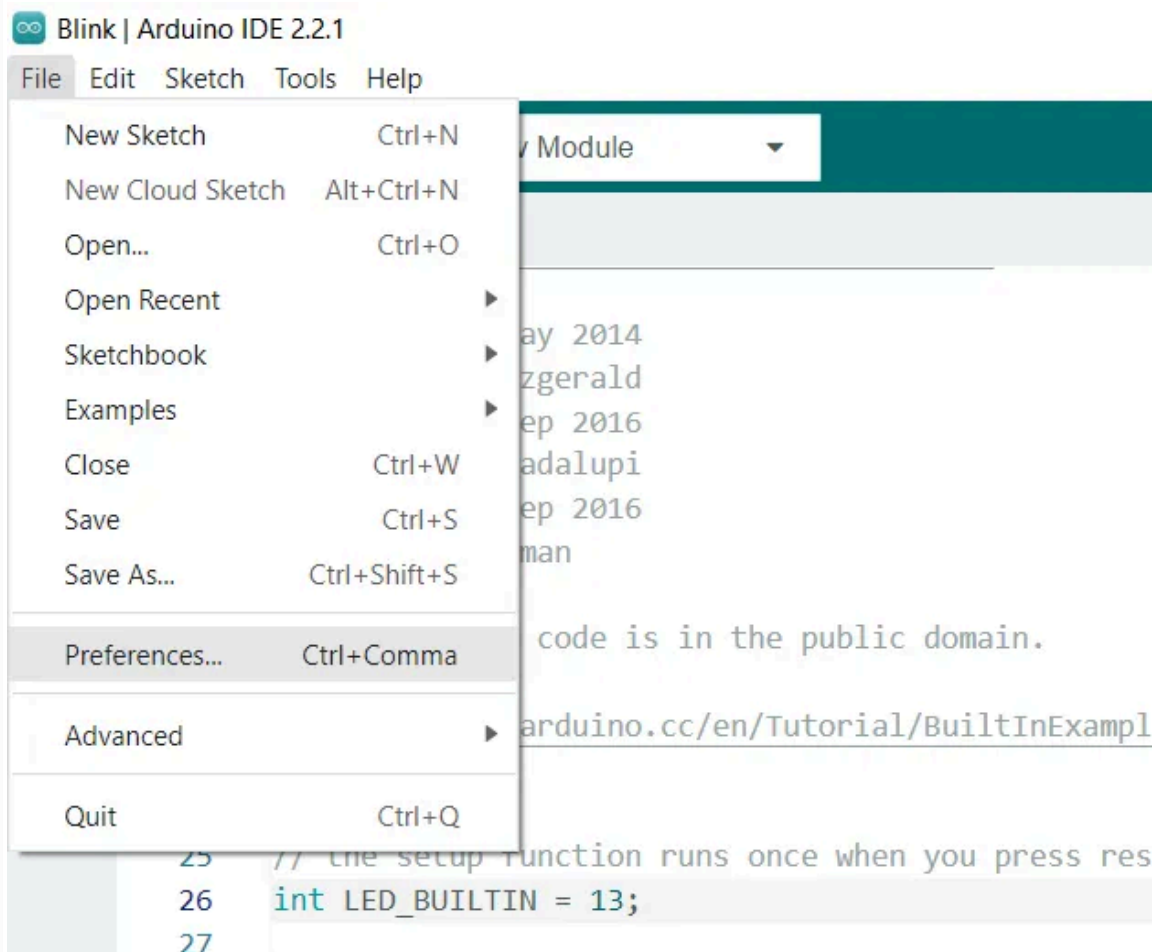
SOURCE CODE
The Arduino IDE 2.0 is open source and its source code is hosted on [GitHub](#).

DOWNLOAD OPTIONS

Windows Win 10 and newer, 64 bits
Windows MSI installer
Windows ZIP file
Linux AppImage 64 bits (X86-64)
Linux ZIP file 64 bits (X86-64)
macOS Intel, 10.14 "Mojave" or newer, 64 bits
macOS Apple Silicon, 11 "Big Sur" or newer, 64 bits
Release Notes

Download the ESP32 Board

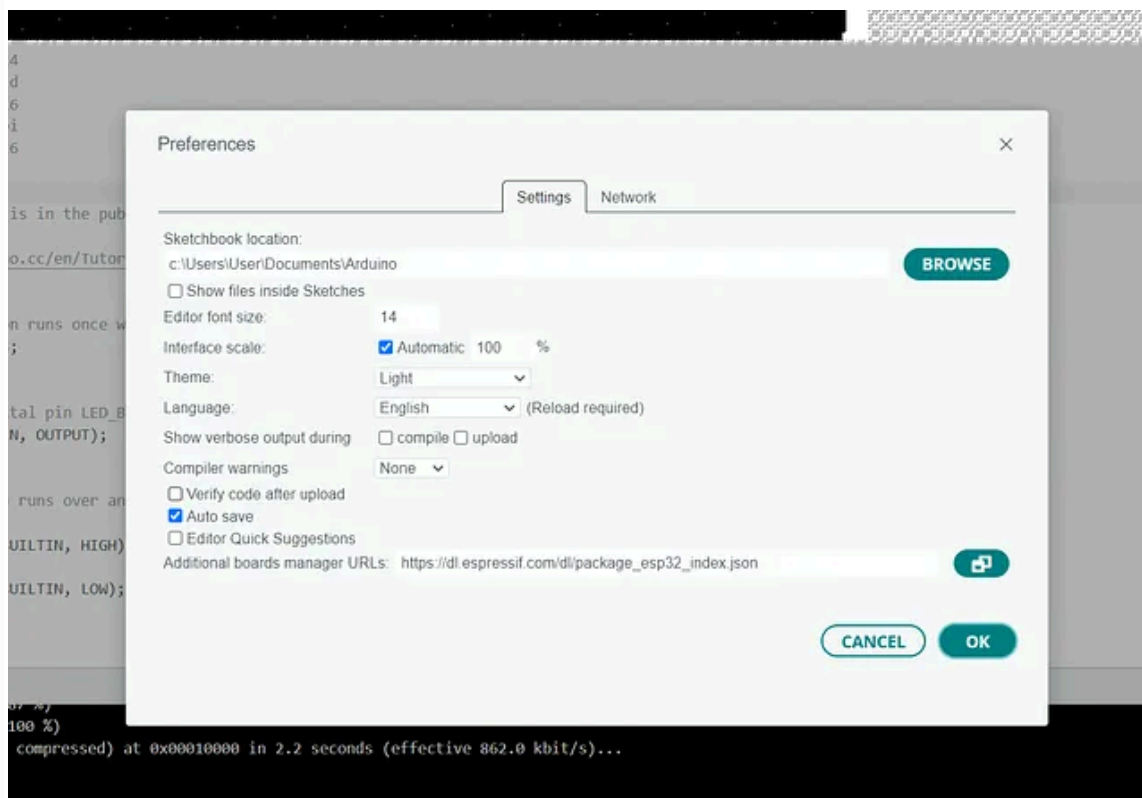
After having installed your Arduino IDE, you will need to install the board on your Arduino IDE so you will be able to select the board. To do this, go to File > Preferences and paste this URL in the additional board manager URL: https://dl.espressif.com/dl/package_esp32_index.json.



```

28 void setup() {
29     // initialize digital pin LED_BUILTIN as an output
30     pinMode(LED_BUILTIN, OUTPUT);
31 }
32
33 // the loop function runs over and over again forever
34 void loop() {

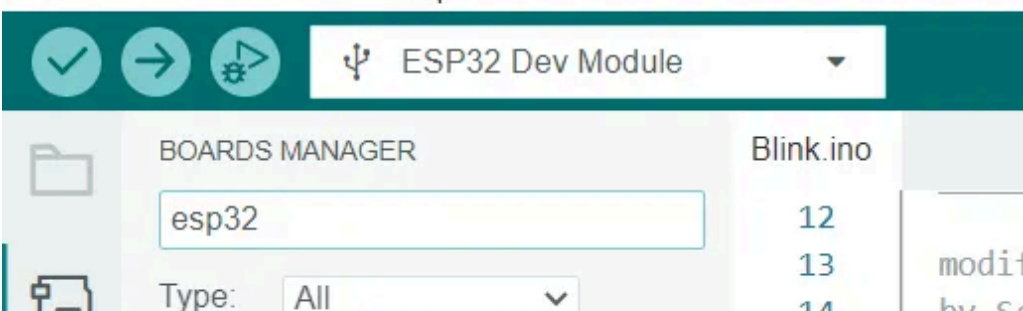
```



Once you are done doing this, go to the board manager on your Arduino IDE and search for ESP32 by espressif. Download the latest version of the board manager, and you are good to go.

Blink | Arduino IDE 2.2.1

File Edit Sketch Tools Help







Arduino ESP32 Boards by Arduino

Boards included in this package:
 Arduino Nano ESP32
[More info](#)

2.0.13 ▾
 INSTALL

esp32 by Espressif Systems

2.0.11 installed

Boards included in this package:
 MGBOT IOTIK 32B, SparkFun ESP32 Thing, u-blox NINA-W10 series...
[More info](#)

2.0.11 ▾
 REMOVE

```

15   modif
16   by Ar
17   modif
18   by Co
19
20   This
21
22   https
23   */
24
25   // the
26   int LED
27
28   void se
29   // in
30   pinMo
31   }
32
33   // the
34   void lo
35   digit
36   delay
37   digit

```

Download the CP210x USB to UART Bridge VCP Drivers

You will need to download the CP210x USB to UART Bridge VCP Drivers for your computer to be able to read the ESP32 board. There are two drivers that ESP32 might be configured to use, but the ESP32-WROOM-32 is configured to use the CP210x USB to UART Bridge VCP Drivers. To download them, kindly go to <https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers?tab=downloads> and select the driver for your computer type.


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// Developers // USB to UART Bridge VCP Drivers

Download and Install VCP Drivers

Downloads for Windows, Macintosh, Linux and Android below.

***Note:** The Linux 3.x.x and 4.x.x version of the driver is maintained in the current Linux 3.x.x and 4.x.x tree at www.kernel.org.

Software Downloads

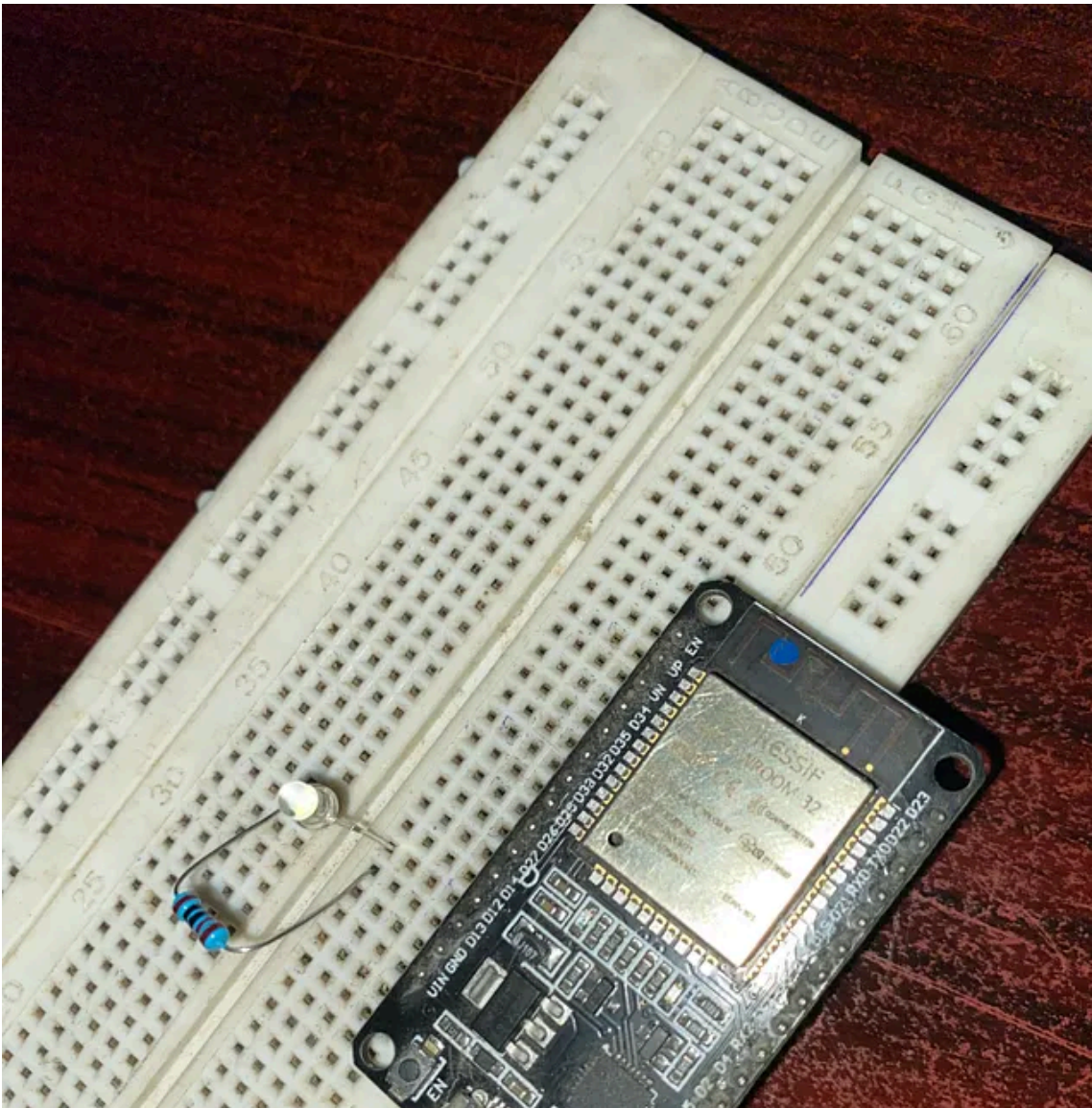
Software (11)
Software · 11

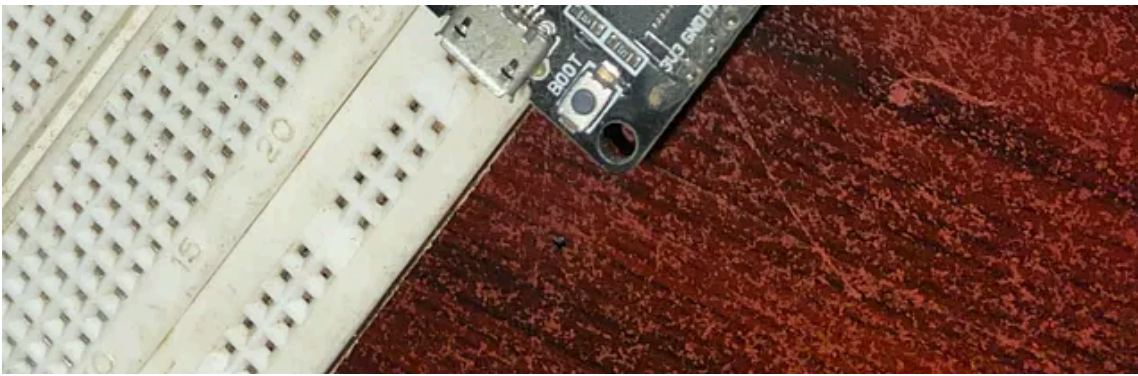
CP210x Universal Windows Driver	v11.3.0 6/24/2023
CP210x VCP Mac OSX Driver	v6.0.2 10/26/2021
CP210x VCP Windows	v6.7 9/3/2020
CP210x Windows Drivers	v6.7.6 9/3/2020
CP210x Windows Drivers with Serial Enumerator	v6.7.6 9/3/2020
CP210x_5x_AppNote_Archive	9/3/2020
CP210x_VCP_Win2K	9/3/2020

NOTE: If you are on some other variant of the ESP32, you might need the [CH340](#) drivers.

Test-run the Board

Once everything is said and done, you are now ready to put the ESP32-WROOM-32 board to the test. Plug in the board to your laptop, and under board selection in your Arduino IDE, select the ESP32 Dev Module as the board. You can upload the blink example after connecting an LED to pin 13 of the ESP board. After compiling, make sure you hold down the boot button on the board so the code will upload.





Congratulations on getting your ESP32-WROOM-Board set up! I am currently running some RTOS tasks on my board; I should write about RTOS later on and how you can set it up on the ESP32. You can follow me to stay updated and get notified whenever that will be, ~~but~~ before then, HAPPY DEVELOPMENT.

Check out my YouTube channel where I make tutorials on how to send data from different development boards to different cloud services like GCP, AWS, Azure, Thingsboard, etc:

https://www.youtube.com/playlist?list=PL6vSC9suLAKNBzQL8pG8CNiiyvkvX_ZUc _____ _____